

ESD Protection in the IHP SG13CMOS5L I/O Library

Structures, test methods, and literature — notes for
`sg13cmos5l_cm_ip__single2diff2single` (Chipalooza 2026)

2 October 2026

Abstract

These notes summarise the electrostatic-discharge (ESD) protection used in the open-source IHP `sg13cmos5l_io` pad library, as read from the CDL netlist and measured in the GDS layout, and explain the background needed to discuss resized clamp transistors with IHP (IHP-Open-PDK issue #1252): the HBM, CDM and TLP test methods, and the grounded-gate NMOS and its relatives. A short annotated bibliography closes the document. Statements marked (*inference*) are inferences from the data, not documented by IHP.

1 Summary

The IHP I/O library uses a conventional, unexotic protection scheme:

- **Primary pad-to-rail diodes** (`DCNDiode` to `iovss`, `DCPDiode` to `iovdd`) on every signal pad.
- An **RC-triggered rail clamp** (a 757 μm -wide thick-oxide NMOS) in every supply pad.
- **Large multi-finger thick-oxide MOSFETs on the pad**: on output pads these are the drivers themselves; on the analog pad they are the same transistor arrays with the gate tied off through a resistor.
- A **secondary stage** (series resistor plus small diodes) in front of anything with a gate connected to the pad.

There are no SCRs, no silicide-block drain ballasting, and the I/O cells do not carry the ESD recognition layer (`Recog.esd`, 99/30).

Origin of the cells. The README of the sibling `sg13g2_io` library states that the cells were generated by the Chips4Makers Python flow (repository `c4m-pdk-ihpsg13g2`, Staf Verhaegen), version 0.0.4, and later patched by IHP (“I/O Update 2”, 18 June 2026, which among other things notes “Analog Pad current capability is improved”, issue #401). The `sg13cmos5l_io` cells have the identical structure. The clamp cells were therefore originally the output of a parametric generator, and Chips4Makers is a second possible point of contact on design intent.

2 Building blocks

Table 1 lists the protection subcircuits in `sg13cmos5l_io.cdl`; Figures 1 and 2 show the analog pad and the rail clamp.

Table 2 shows which device sits on which pad. All pad clamps use the same finger cell; only the gate connection differs. The analog pad’s 20-finger pair is larger than even the 30 mA driver.

Table 1: ESD building blocks in `sg13cmos51_io`.

Cell	What it is	Used in
DCNDiode, DCPDiode	Primary diodes: n+/p-sub (dantenna) from iovss to pad, p+/n-well (dpantenna) from pad to iovdd. Each is two $1.26\text{ }\mu\text{m} \times 27.78\text{ }\mu\text{m}$ stripes.	every signal pad; supply-ground pads
Clamp_N43N43D4R, RCCLampResistor, RCCLampInverter	Rail clamp: 172 fingers of $4.4\text{ }\mu\text{m}/0.6\text{ }\mu\text{m}$ sg13_hv_nmos ($757\text{ }\mu\text{m}$ total). Trigger: $26 \times 5.24\text{ k}\Omega$ rppd ($136\text{ k}\Omega$) into a $\approx 1200\text{ }\mu\text{m}^2$ thick-oxide MOS capacitor; inverter (PMOS $350\text{ }\mu\text{m}/0.5\text{ }\mu\text{m}$, NMOS $2 \times 54\text{ }\mu\text{m}/0.5\text{ }\mu\text{m}$) drives the gate.	IOPadVdd, IOPadIOVdd
Clamp_{N,P} {2,8,15}N...D	Output drivers that double as the ESD device; gate is a pin, driven by GateLevelUpInv .	IOPadOut, TriOut, InOut (4/16/30 mA)
Clamp_N20NOD, Clamp_P20NOD	Same arrays with gate tied off through rppd: $1.96\text{ k}\Omega$ to iovss (gate-coupled NMOS) and $6.77\text{ k}\Omega$ to iovdd (gate-to-VDD PMOS).	IOPadAnalog only
SecondaryProtection	$587\text{ }\Omega$ rppd series resistor plus small dantenna / dpantenna diodes to the rails.	IOPadAnalog (padres), LevelDown

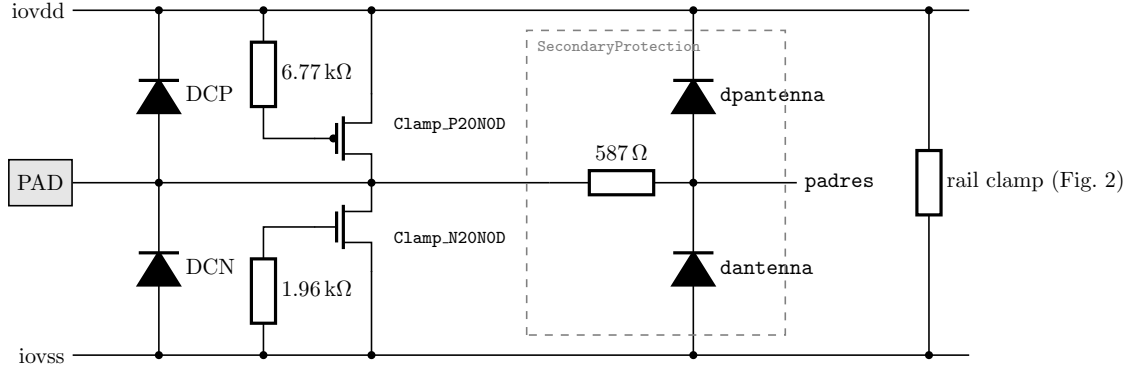


Figure 1: ESD network of `sg13cmos51_IOPadAnalog` (from the CDL). The core connects to **padres**, behind the $587\text{ }\Omega$ secondary resistor. NMOS bulk is the substrate (tied to iovss through ptap guard rings), PMOS bulk is iovdd. The rail clamp is not part of this pad; it sits in the IOPadVdd/IOPadIOVdd supply pads and completes the pad $\rightarrow$ iovdd $\rightarrow$ iovss discharge path.

3 Clamp geometry (measured from the GDS)

The following was measured with the KLayout Python API on `sg13cmos51_io.gds` from the IHP-Open-PDK dev branch.

- Devices: **sg13_hv_nmos** and **sg13_hv_pmos**, $L = 0.6\text{ }\mu\text{m}$. Finger width $4.4\text{ }\mu\text{m}$ (NMOS) and $6.66\text{ }\mu\text{m}$ (PMOS). PMOS cells contain two rows, so P20 has 40 gates.
- Finger pitch alternates $1.78\text{ }\mu\text{m} / 1.24\text{ }\mu\text{m}$. Contact-to-gate spacing is $0.51\text{ }\mu\text{m}$ on the shared drains (confirmed by net tracing: the wide strips connect to **pad**) and $0.24\text{ }\mu\text{m}$ on the shared sources (iovss/iovdd). This extra drain spacing is the only ballasting.
- Drains are fully silicided: **SalBlock** appears only on the rppd gate resistors of the NOD variants.
- No **Recog.esd** (99/30) layer: the clamps extract and pass LVS as ordinary HV MOSFETs. Each cell has its own ptap guard ring.
- Cell width $80.4\text{ }\mu\text{m}$ – $81.2\text{ }\mu\text{m}$, matching the pad pitch.

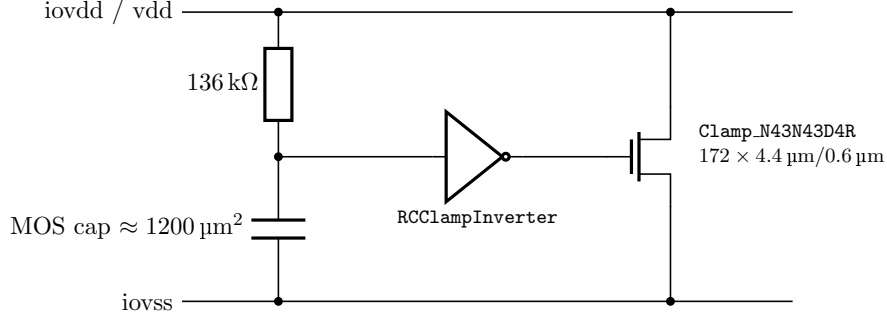


Figure 2: RC-triggered rail clamp in IOPadVdd and IOPadIOVdd (inverter supplied from the protected rail and iovss). On a fast ESD edge the capacitor holds the inverter input low, the inverter drives the clamp gate high, and the clamp conducts through its channel — no snapback is involved. On a slow power-up the input follows the rail and the clamp stays off. With a thick-oxide C_{ox} of roughly $5 \text{ fF } \mu\text{m}^{-2}$ the time constant is on the order of $1 \mu\text{s}$ (*inference*; C_{ox} not taken from IHP data).

Table 2: Which protection device sits on which pad (from `sg13cmos5l_io.cdl`).

Pad	Pad $\rightarrow$ iovss	Pad $\rightarrow$ iovdd	Other
IOPad{Out,TriOut,InOut}4mA	Clamp_N2N2D (driver)	Clamp_P2N2D (driver)	DCN/DCP diodes
IOPad{Out,TriOut,InOut}16mA	Clamp_N8N8D (driver)	Clamp_P8N8D (driver)	DCN/DCP diodes
IOPad{Out,TriOut,InOut}30mA	Clamp_N15N15D (driver)	Clamp_P15N15D (driver)	DCN/DCP diodes
IOPadAnalog	Clamp_N20N0D (GCNMOS)	Clamp_P20N0D (GDPMOS)	diodes, secondary stage
IOPadIn	—	—	diodes; secondary stage inside LevelDown
IOPadVdd, IOPadIOVdd	Clamp_N43N43D4R, RC-triggered rail clamp		
IOPadVss, IOPadIOVss	—		DCN/DCP diodes

- Naming (*inference*): `Clamp_{N,P}aNbD[rR]` reads as a fingers per row, b of them with a driven gate ($b = 0$: gate tied off), and r rows (4R: $43 \times 4 = 172$ fingers).

4 Standalone ESD devices in SG13_dev

The PCell `esd_code.py` offers `diodevdd_2kv`, `diodevdd_4kv`, `diodevss_2kv`, `diodevss_4kv`, `nmosc1_2` and `nmosc1_4`. They are fixed layouts (the only parameter is the model name), drawn with `Recog.esd` plus a text label, which the DRC deck uses to exempt them from antenna, well-crossing and salicide-block-resistor rules. They are the only place in the open PDK that states a kV rating. The I/O library does not use them. The DRC deck also recognises an `scr1` device, for which CMOS5L ships no PCell.

5 ESD test methods: HBM, CDM and TLP

HBM (Human Body Model). Models a charged person touching a pin: a 100 pF capacitor charged to the test voltage and discharged into the pin through $1.5 \text{ k}\Omega$, giving a pulse with a rise time of a few nanoseconds and a decay time constant of about 150 ns . Peak current is approximately $V/1.5 \text{ k}\Omega$, so 2 kV HBM corresponds to about 1.3 A . Standardised in ANSI/ESDA/JEDEC JS-001; performed on packaged parts over pin combinations, with a pass/fail voltage as the result.

CDM (Charged Device Model). The device itself is charged and discharges through a single pin: a pulse of about 1 ns with several amperes peak. Standardised in ANSI/ESDA/JEDEC JS-002. CDM is the threat a secondary stage such as **SecondaryProtection** primarily addresses, since the charge resides on the die and stresses the thin input gates directly.

TLP (Transmission Line Pulse). A characterisation method, not a qualification test. A charged length of coaxial line is discharged into the device under test, producing a clean rectangular pulse, typically 100 ns wide, comparable in energy to an HBM event. The amplitude is stepped; at each step voltage and current are averaged over the flat part of the pulse, and a DC leakage measurement after each pulse detects damage. The result is the I–V curve in the ESD regime (Fig. 3). Standardised in ANSI/ESD STM5.5.1; the 1 ns–5 ns variant (vf-TLP) correlates with CDM.

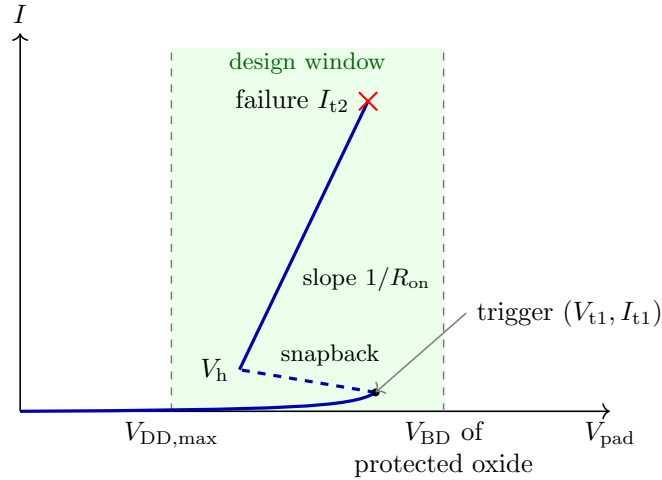


Figure 3: Qualitative TLP I–V curve of a snapback device (GGNMOS); illustrative, not IHP data. V_{t1} must lie inside the design window; V_h below $V_{DD,max}$ risks a latched-on clamp in normal operation; I_{t2} sets the robustness, with the rule of thumb HBM rating $\approx I_{t2} \times 1.5 \text{ k}\Omega$.

Why it matters here. I_{t2} is normally quoted per micrometre of gate width for a given finger design (L, drain contact-to-gate spacing, silicidation). With TLP data for IHP’s $4.4 \mu\text{m}/0.6 \mu\text{m}$ finger, choosing a finger count is arithmetic, subject to uniform triggering of all fingers. Changing the finger geometry itself invalidates any per-micrometre figure. The most specific question to put to IHP is therefore whether TLP curves exist for the **Clamp_*** finger cells.

6 GGNMOS and related devices

GGNMOS (grounded-gate NMOS). Drain to the pad; gate, source and bulk to VSS. The device is off in normal operation and protects through its parasitic lateral NPN (Fig. 4). On a positive zap the drain–substrate junction avalanches at V_{t1} ; avalanche holes flowing through R_{sub} to the ptap raise the local substrate potential until the source junction forward-biases; the NPN turns on and the voltage snaps back to V_h , after which the device carries amperes until it fails at I_{t2} . On a negative zap the drain–substrate diode simply conducts forward. The known weakness is multi-finger triggering: the first finger to snap back pulls the pad below the others’ V_{t1} , and if $V_{t2} < V_{t1}$ that single finger carries all the current and fails. Remedies are drain ballast resistance (silicide block, larger drain contact spacing) and gate coupling.

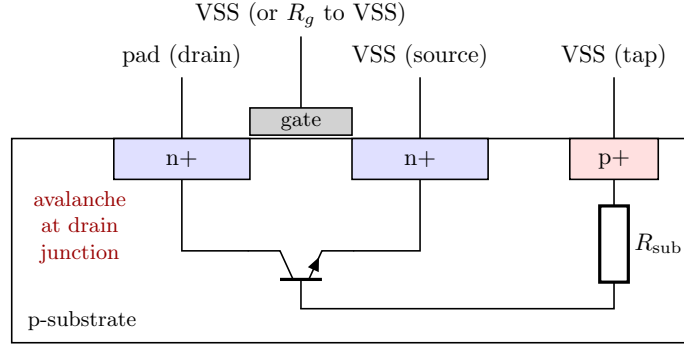


Figure 4: Cross-section of a grounded-gate NMOS with its parasitic lateral NPN (collector = drain, base = substrate, emitter = source) and the substrate resistance R_{sub} to the ptap.

GCNMOS (gate-coupled NMOS). The gate is tied to VSS through a resistor rather than directly. The fast ESD edge couples onto the gate through the gate-drain overlap capacitance; the channel conducts briefly, which lowers V_{t1} and makes the fingers trigger together. IHP's `Clamp_N20N0D` (1.96 k Ω to iovss) is of this kind.

GDPMOS (gate-to-VDD PMOS). The mirror image: drain to pad; source, gate and n-well to VDD. The parasitic PNP snaps back weakly or not at all in modern processes, so the device mainly acts as a p+/n-well diode from pad to VDD plus a breakdown path. IHP's `Clamp_P20N0D` (6.77 k Ω to iovdd) is of this kind.

Self-protecting output driver. An output NMOS/PMOS large enough to serve as its own clamp. During an ESD event on an unpowered chip the pre-driver does not actively drive the gates, so the drivers behave roughly like GGNMOS/GDPMOS. All IHP output pads use this scheme.

RC-triggered (active) rail clamp. A large MOSFET that conducts through its channel, with the gate driven by an RC transient detector; no snapback is required (Fig. 2).

SCR / LVTSCR. Thyristor-based clamps: very area-efficient but latch-up-prone. Not used in the IHP I/O library.

7 Implications for resized clamp PCells

- **DRC and LVS do not check ESD robustness.** The only ESD-related checks in the CMOS5L layout rules are the latch-up rules LU.a–LU.d (§7.2, off by default), together with guidelines for output buffers: guard rings around every pad-connected device, and double guard rings between N and P drivers and between drivers and core. Enabling the LU check is worthwhile.
- **Conservative path:** keep the finger cell identical (L, W, drain spacing, contact pattern, guard ring) and change only the finger count. Robustness then scales roughly with total width, provided all fingers trigger uniformly; with silicided drains, uniform triggering is the weak point, which is what the gate-coupling resistors of the N0D variants address.
- **Argument for IHP:** an output driver at least as wide as N8N8D/P8N8D, alongside the DCN/DCP diodes, is structurally the same as IHP's own 16 mA output pad.
- **What only IHP can supply:** TLP/HBM data for these cells. No ESD rating for the I/O library was found in the open PDK.

8 Observed netlist defect

In `sg13cmos51_IOPadInOut30mA` the instance

```
XI3 net2 iovss pad ptap1 / sg13cmos51_Clamp_N15N15D
```

passes four nodes to a three-port subcircuit. A SPICE simulator should reject this; it may merit a separate one-line issue.

9 Literature

Books

- [B1] A. Amerasekera and C. Duvvury, *ESD in Silicon Integrated Circuits*, 2nd ed. Wiley, 2002. — The standard reference on device physics (snapback, second breakdown, silicide effects) and protection design.
- [B2] S. Dabral and T. J. Maloney, *Basic ESD and I/O Design*. Wiley, 1998. — Written from the I/O designer’s perspective; closest in spirit to the pad work here.
- [B3] S. H. Voldman, *ESD: Physics and Devices*, Wiley, 2004; and *ESD: Circuits and Devices*, Wiley, 2006 (2nd ed. 2015). — Broad series; the circuits volume covers rail clamps and I/O networks.
- [B4] A. Z. H. Wang, *On-Chip ESD Protection for Integrated Circuits: An IC Design Perspective*. Kluwer, 2002.
- [B5] O. Semenov, H. Sarbishaei and M. Sachdev, *ESD Protection Device and Circuit Design for Advanced CMOS Technologies*. Springer, 2008. — Has dedicated chapters on circuit concepts and on power (rail) clamps.

Foundational papers

- [P1] T. J. Maloney and N. Khurana, “Transmission line pulsing techniques for circuit modeling of ESD phenomena,” *Proc. EOS/ESD Symp.*, EOS-7, pp. 49–54, 1985. — Origin of TLP.
- [P2] C. H. Polgreen and A. Chatterjee, “Improving the ESD failure threshold of silicided n-MOS output transistors by ensuring uniform current flow,” *IEEE Trans. Electron Devices*, vol. 39, no. 2, 1992. (*volume/issue not independently verified*) — Directly addresses the multi-finger uniformity problem of silicided drivers like IHP’s.
- [P3] R. Merrill and E. Issaq, “ESD design methodology,” *Proc. EOS/ESD Symp.*, EOS-15, pp. 233–237, 1993. — Early statement of the rail-based (diodes plus power clamp) protection scheme.
- [P4] E. R. Worley *et al.*, “Sub-micron chip ESD protection schemes which avoid avalanching junctions,” *Proc. EOS/ESD Symp.*, EOS-17, pp. 13–20, 1995. — The RC-triggered MOSFET rail clamp.
- [P5] J. Z. Chen, E. A. Amerasekera and C. Duvvury, “Design methodology and optimization of gate-driven NMOS ESD protection circuits in submicron CMOS processes,” *IEEE Trans. Electron Devices*, vol. 45, no. 12, pp. 2448–2456, 1998, doi:10.1109/16.735721. — Gate coupling (GCNMOS) for uniform multi-finger turn-on.
- [P6] M.-D. Ker, “Whole-chip ESD protection design with efficient VDD-to-VSS ESD clamp circuits for submicron CMOS VLSI,” *IEEE Trans. Electron Devices*, vol. 46, no. 1, pp. 173–183, 1999.

Freely available

- [F1] J.-H. Chun, *ESD Protection Circuits for Advanced CMOS Technologies*, Ph.D. dissertation, Stanford University, 2006. <http://www-tcad.stanford.edu/tcad/pubs/theses/chun.pdf>
- [F2] Industry Council on ESD Target Levels, *White Paper 1: A Case for Lowering Component Level HBM/MM ESD Specifications and Requirements*, and *White Paper 2: A Case for Lowering Component Level CDM ESD Specifications and Requirements* (Rev. 3.0, 2021). <https://www.esdindustrycouncil.org/ic/en/documents> — Why 1 kV HBM / 250 V CDM-class targets are considered safe with basic factory ESD control.
- [F3] ESDA / In Compliance, “Characterization for ESD Design, the TLP Zoo” (article series). <https://www.esda.org/news/characterization-for-esd-design-the-tlp-zoo-part-2/>
- [F4] “Transmission Line Pulse Testing: The Indispensable Tool for ESD Characterization of Devices, Circuits and Systems,” *In Compliance Magazine*. <https://incompliancemag.com/transmission-line-pulse-testing-the-indispensable-tool-for-esd-characterization-of-devices-circuits-and-systems/>
- [F5] M.-D. Ker’s ESD reference list (NYCU). <https://alab.iee.nycu.edu.tw/~mdker/ESD/refer/refer.html>

Standards

ANSI/ESDA/JEDEC JS-001 (HBM); ANSI/ESDA/JEDEC JS-002 (CDM); ANSI/ESD STM5.5.1 (TLP); ANSI/ESD SP5.5.2 (vf-TLP).

PDK sources used for Sections 1–7

- IHP-Open-PDK (dev branch): `ihp-sg13cmos5l/libs.ref/sg13cmos5l_io/cdl/sg13cmos5l_io.cdl`, `.../gds/sg13cmos5l_io.gds`, `ihp-sg13g2/libs.ref/sg13g2_io/doc/README.md`, `ihp-sg13cmos5l/libs.doc/doc/SG13CMOS5L_os_layout_rules.pdf` (§7.2), `.../sg13cmos5l_pycell_lib/ihp/esd_code.py`. <https://github.com/IHP-GmbH/IHP-Open-PDK>
- IHP-Open-PDK issue #1252. <https://github.com/IHP-GmbH/IHP-Open-PDK/issues/1252>
- Chips4Makers, `c4m-pdk-ihpsg13g2`. <https://gitlab.com/Chips4Makers/c4m-pdk-ihpsg13g2>